KIMATHI KAAI

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EDUCATION

Doctor of Philosophy (Ph.D.)

University of Waterloo

Systems Design Engineering, Class of 2028

Advisors: Prof. Alexander Wong, Asst. Prof Sirisha Rambhatla

Master of Applied Science (M.A.Sc.)

University of Waterloo

Systems Design Engineering, Class of 2024

Thesis: *Addressing Data Scarcity in Domain Generalization for Computer Vision Applications in Image Classification*

Advisors: Prof. Alexander Wong, Asst. Prof Sirisha Rambhatla

AWARDS AND HONORS

IBET Momentum Fellowship Award (2024)

4th Year Design Project I-Beam Award (2022)

President's Scholarship of Distinction (2017)

PUBLICATIONS/WORKSHOPS

J. Park, K. Kaai, S. Hossain, N. Sumi, S. Rambhatla, P. Fieguth. **Domain-Guided Spatio-Temporal Self-Attention for Egocentric 3D Pose Estimation.** ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2023.

K. Kaai, S. Hossain, S. Rambhatla. **Are all classes created equal? Domain Generalization for Domain-Linked Classes.** NeurIPS Distribution Shifts Workshop, 2023.

RESEARCH FOCUS

Machine Learning, Computer Vision, Domain Generalization/Adaptation, Continuous Learning, Self-Supervised Learning, Semantic Segmentation, Vision-Language Modelling

TECHNICAL SKILLS

Python, Git, Pytorch, TensorFlow, Linux

RESEARCH EXPERIENCE

Graduate Researcher

University of Waterloo | 09/2022 - Present

- Advanced domain generalization computer vision techniques, demonstrating expertise in reducing model dependency on labelled data; presented this research at NeurIPS 2023
- Co-designed ego-centric 3D human pose estimation models in collaboration with **Nissan**, significantly enhancing training tools for factory environments
- Addressed data scarcity challenges at **Apple**, reducing dependency on annotated datasets in visual defect detection

INDUSTRY EXPERIENCE

Deep Learning Developer

Applied Brain Research | 04/2021 - 09/2021

- Designed recurrent neural network architectures for wearable and computer vision applications. Focus was on human activity recognition from human electrocardiogram signals

Vision Software Engineer

Cepton Technologies | 06/2020 - 04/2021

- Optimized point-cloud surface feature descriptors for LiDAR-based human/vehicle recognition with C++11, PCL and OpenCV
- Researched and prototyped LiDAR-based ego-motion vehicle estimation models on the Automotive Perception Team

Hardware Designer

Evertz Microsystems | 09/2019 - 12/2019

- Developed multi-channel J2K encoding/decoding products for Amazon and ESPNs streaming services using C/C++
- Designed data corruption detection modules for video and audio ST2110-20/30 IP packets in Python
- Collaborated with product specialists to perform end-to-end integration tests for the video and audio encoders

Software Engineer

Vena Solutions | 01/2019 - 04/2019

- Developed navigation wizards to integrate Vena's FP&A platform with external data sources using React and Redux
- Debugged the Java 8 back-end to diagnose issues with MongoDB and SQL Server transactions